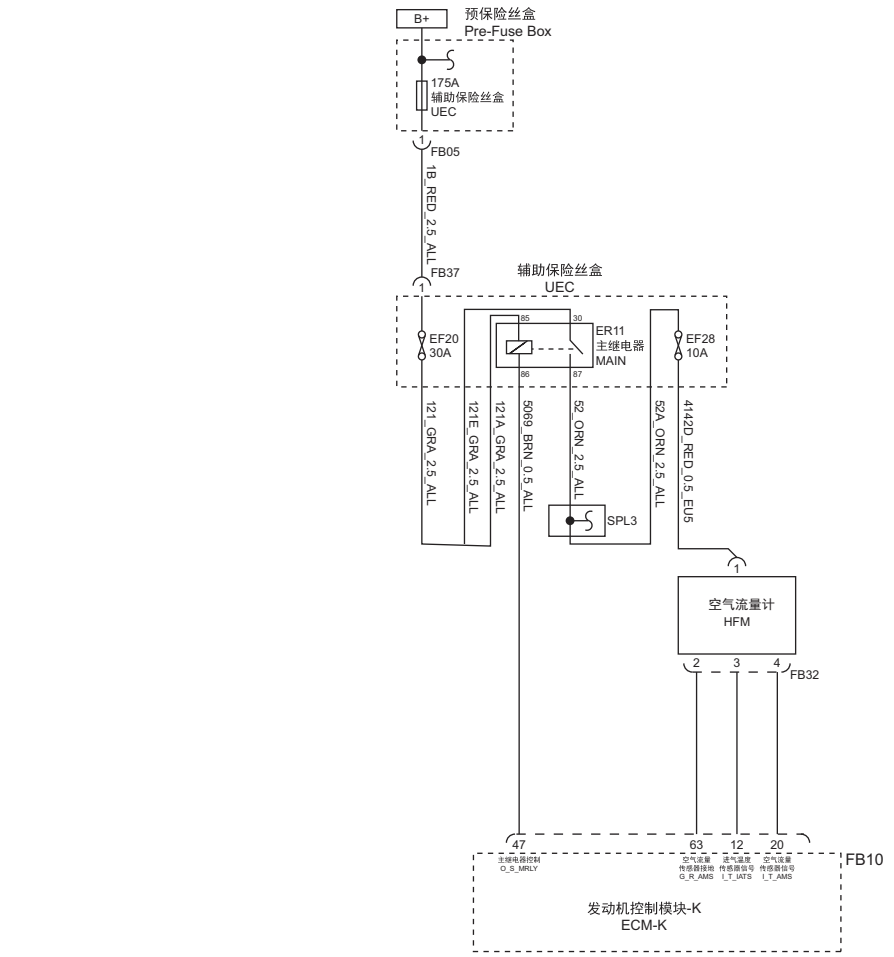
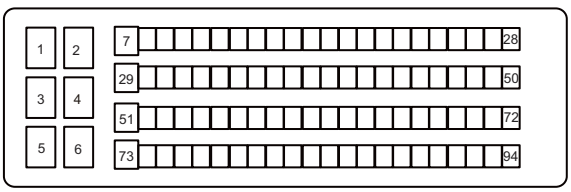
P0112、P0113

Description of fault diagnosis codes
P0112-11：SRC low for air temperature sensor of HFM
P0112-36：SRC low for period duration of air temperature sensor of HFM
P0113-12：SRC high for air temperature sensor of HFM
P0113-37：SRC high for period duration of air temperature sensor of HFM
Detection tools
1 Multimeter and diagnostic apparatus
Possible causes
• Fault in harness or harness plug • Inlet air temperature sensor (build-in the air flow meter) fault
Description of circuit diagram
The intake air temperature (IAT) sensor is a variable resistor, which is the constituent element of mass air flow/intake air temperature (MAF/IAT) sensor. The intake air temperature can change the voltage on 5v signal circuit provided by the engine control module (ECM). The signal varies with the intake air temperature inside the mass air flow (MAF) sensor hole, and it is displayed as °C (°F) on the fault diagnosis instrument. The engine control module provides reference voltage circuit of low level to the sensor.

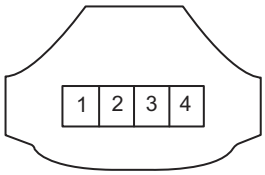
Circuit diagram



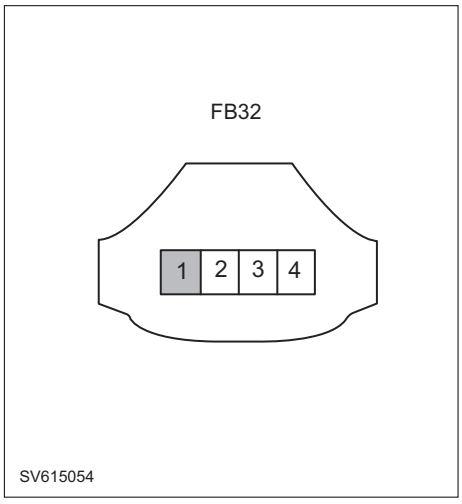
FB10

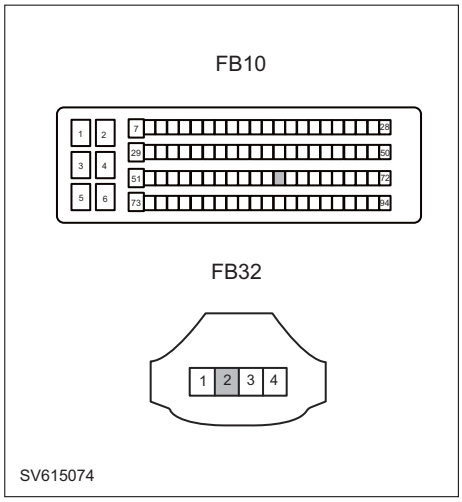


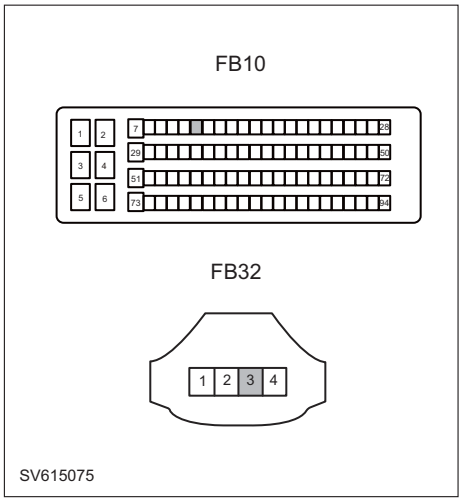
FB32



SV615014

Troubleshooting procedures		
Step 1	General inspection.	
		1 Check the fuse for damage. 2 Check battery capacity. 3 Check the engine control module and the harness plug of air flow meter for damage, poor contact, aging and looseness. 4 Check whether the result is normal.
		No → Repair or replace the faulted parts.
No ↓		
Step 2	Check fault codes.	
		1 Connect the fault diagnosis instrument to diagnosis interface. 2 Turn the ignition switch to ON position. 3 Diagnose the engine control system with diagnostic apparatus. 4 Is there any other fault code except for P0112,P0113?
		Yes → Refer to:DTC Summary List (EU5).
No ↓		
Step 3	Check the power supply of air temperature sensor.	
		1 Turn the ignition switch to OFF. 2 Disconnect the harness plug of air flow meter. 3 Turn the ignition switch to ON position. 4 Check the voltage between the harness plug of the air flow meter and ground. Measure circuit: the voltage between FB32 plug terminal 1 and ground. **Standard value: 11~14V** 5 Check whether the result is normal?
		No → Repair or replace the faulted parts.
Yes ↓		

Step 4	Check the ground circuit of air temperature sensor (open circuit).
	1 Turn the ignition switch to OFF. 2 Disconnect the harness plug of engine control module. 3 Check the resistance of harness plug between air flow meter and engine control module. Measure circuit: the resistance between FB32 plug terminal 2 and FB10 plug terminal 63. **Standard value: ≈ 0Ω** 4 Check whether the result is normal?
	No → Repair or replace the faulted parts.
Yes ↓	
Step 5	Check the ground circuit of air temperature sensor (short circuit).
	1 Check the resistance between the harness plug of air flow meter and ground. Measure circuit: the resistance between FB32 plug terminal 2 and ground. **Standard value: ∞** 2 Check the voltage between the harness plug of the air flow meter and ground. Measure circuit: the voltage between FB32 plug terminal 2 and ground. **Standard value: ≈ 0V** 3 Check whether the result is normal?
	No → Repair or replace the faulted parts.
Yes ↓	

Step 6	Check the signal circuit of air temperature sensor (open circuit).
	1 Check the resistance between the harness plug of air flow meter and the plug of engine control unit. Measure circuit: the resistance between FB32 plug terminal 3 and FB10 plug terminal 12. **Standard value: ≈ 0Ω** 2 Check whether the result is normal?
	No → Repair or replace the faulted parts.
Yes ↓	
Step 7	Check the signal circuit of air temperature sensor (short circuit).
	1 Check the resistance between the harness plug of air flow meter and ground. Measure circuit: the resistance between FB32 plug terminal 3 and ground. **Standard value: ∞** 2 Check the voltage between the harness plug of the air flow meter and ground. Measure circuit: the voltage between FB32 plug terminal 3 and ground. **Standard value: ≈ 0V** 3 Check whether the result is normal?
	No → Repair or replace the faulted parts.
Yes ↓	
Step 8	Check the air temperature sensor(components).
	1 Check the resistance among the terminals of air flow meter. Measure circuit: the resistance between terminal 1 and terminal 2 of air flow meter. **Standard value: ≈ 17.5KΩ(13.5 ℃)** 2 Check whether the result is normal?
	No → Replace the air flow meter (build-in air temperature sensor).
Next ↓	
Step 9	Complete the check and confirm whether the system works normally nor not.